

Alec Brooks

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Summary

Data Science student and casino surveillance professional with experience in statistical analysis, research data workflows, operational analytics, and dashboard development. Co-author of a peer-reviewed ecological research publication with internship experience in data science research at the University of Nevada, Las Vegas. Skilled in SQL, Python, R, VBA, C++, Excel, Tableau, and Power BI.



Skills

Data analysis, hypothesis testing, and statistical reasoning
Programming and query languages: SQL, Python, R, VBA, C++
Data visualization and dashboarding: Tableau, Power BI, Excel
Workflow automation, data cleaning, and operational analytics



Research & Publications

EcoBlitz Flowering Phenology Research & Publication

2024 – 2026, Truckee Meadows Community College

- Co-author of peer-reviewed ecological research published in Ecology and Evolution.
- Contributed to statistical analysis of time-series flowering survey data in a Great Basin Desert ecosystem.
- Presented research findings at data science and undergraduate research conferences, including the University of Nevada, Reno; INBRE Anchorage, Alaska; and INBRE Las Vegas.
- Lahti, M. E., Sakamura, E., Jung, M., Vigil, C. M., Smith, K. S., Ramirez, M., & Brooks, A. C. (2026). Rapid surveys reveal temporal variation in flowering community phenology in a Great Basin Desert ecosystem. Ecology and Evolution, 16, e73340. <https://doi.org/10.1002/ece3.73340>

UNLV Carbonaceous Aerosols Speciation Analysis Internship

Data Science Intern, University of Nevada, Las Vegas 2025

- Developed Python and SQL workflows to aggregate, clean, and store longitudinal air-quality instrument data.
 - Built a SQLite database structure for organizing AE33 and TCA instrument outputs across long-term monitoring periods.
 - Performed time-series analysis on carbonaceous aerosol speciation data, including timestamp alignment and missing-data handling.
 - Presented research findings at a University of Nevada, Reno data science conference.
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Professional Experience

Peppermill Resort, Spa and Casino / Surveillance Agent

December 2018 – Present, Reno, NV

- Built Excel and VBA-based tools to analyze player behavior and identify potential advantage play, cheating, and card counting.
- Used point-of-sale logging systems and transaction reports to investigate suspicious activity, including voids, cancels, refunds, discounts, and no-sale events.
- Developed operational dashboards and reports to detect irregular transaction patterns by employee, terminal, time of day, and transaction type.
- Applied statistical reasoning and hypothesis testing to evaluate surveillance patterns, reduce false positives, and support evidence-based investigations.



Projects

CASS-DB-Manager

Python, SQLite, Data Engineering

- Developed a Python application for storing, managing, and analyzing AE33 Aethalometer air-quality data during a UNLV Data Science internship.
- Designed database workflows for organizing longitudinal aerosol black carbon measurements used in environmental research.
- Implemented automated data storage and retrieval processes to support monitoring and aerosol speciation analysis.
- <https://github.com/AlecBrooks/CASS-DB-Manager>

ChessPI

R, APIs, Data Acquisition

- Developed an open-source R package for downloading and analyzing player game records through the Chess.com API.
- Implemented functions for retrieving historical game archives and collecting games across user-defined date ranges.
- Automated acquisition of real-world chess data for statistical analysis and performance evaluation.
- <https://github.com/AlecBrooks/ChessPI>



Education

Truckee Meadows Community College / Data Science

August 2023 – Present, Reno, NV

Currently enrolled in a Data Science bachelor's degree transfer program.

Awarded Outstanding Student Award, Data Science / Computer Science Division, 2026

NCLab / Data Literacy Career Certificate

June 2024 – August 2024, Online

Completed an accelerated Data Science training program with hands-on experience in data visualization, statistical analysis, and hypothesis testing. Demonstrated proficiency in Google Sheets, Microsoft Excel, and Power BI for creating dynamic dashboards and data-driven insights.

Google / Data Analytics Professional Certificate

February 2023 – August 2023, Online

Completed a six-month career certificate covering data cleaning, data visualization, project management, databases, and communicating analytics findings. Developed practical experience transforming complex data into actionable insights.